

PishieGen: A Biomimetic Artificial-Life Simulator for Genotype–Phenotype Mapping, Heritable Variation, and Environment-Dependent Fitness

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Abstract

Artificial-life systems provide a controlled way to study how simple local rules, inherited variation, and environmental constraints may generate population-level adaptive dynamics. This paper introduces *PishieGen*, a biomimetic artificial-life simulator designed around a compact digital genome, explicit genotype–phenotype mapping, rule-based inheritance, stochastic mutation, and environment-dependent fitness. The framework is motivated by three requirements common to computational models of adaptive biological systems: (i) traits should be represented in a way that separates genotype from phenotype, (ii) inheritance and mutation should be auditable and experimentally testable, and (iii) organism performance should depend on interactions between expressed traits and ecological context rather than on a single global score. *PishieGen* models digital organisms as agents with fixed-length binary genomes that encode morphological, physiological, sensory, health-related, and behavioral traits. Phenotypes are expressed through deterministic and stochastic decoding functions that support dominance, recessivity, polygenic aggregation, pleiotropic constraints, and lethal or deleterious genotype combinations. Environments are represented as biome objects with climate, substrate, vegetation, illumination, and resource parameters. Fitness is computed as a transparent composition of thermal compatibility, camouflage, locomotor suitability, sensory advantage, health burden, and behavioral timing. The proposed system is not intended as a biological replica of any species. Instead, it is a reproducible simulation platform for studying stylized questions about artificial inheritance, biomimetic design abstraction, selective breeding, mutation-driven diversity, and genotype–environment interaction. We specify the mathematical model, software architecture, validation plan, and initial experimental protocols suitable for open-source implementation and reproducible computational experiments.

Keywords: artificial life; biomimicry; genotype–phenotype map; agent-based modeling; individual-based ecology; digital organisms; evolutionary simulation; computational biology.

1 Introduction

Artificial life studies synthetic systems that exhibit life-like organization or behavior in artificial media, including computational environments [1]. Within this tradition, simulation is not merely a visualization tool; it is a method for asking how population-level dynamics may arise from local rules, heritable variation, and interactions between agents and environments. Several open problems in artificial life concern evolvability, emergence, adaptive complexity, and the construction of systems capable of producing sustained novelty rather than short-lived optimization [2, 3]. A major challenge is that many artificial evolutionary systems compress biological complexity into simple scalar fitness functions, making it difficult to inspect how inherited structures produce phenotypes and how phenotypes interact with ecological constraints.

The genotype–phenotype map is a central unresolved problem in evolutionary theory and artificial-life modeling. Modern reviews emphasize that understanding how genotypes map onto molecular phenotypes, organismal function, and fitness remains a cornerstone for predictive evolutionary dynamics [4]. In computational systems, the genotype–phenotype map is not constrained by biochemistry in the same way as natural genomes; however, this freedom creates a methodological responsibility. If the mapping is arbitrary or opaque, simulation results become difficult to interpret. If the mapping is explicit, modular, and experimentally testable, the system can serve as a useful artificial laboratory for studying stylized evolutionary mechanisms.

PishieGen is proposed as such a laboratory. The simulator represents digital organisms using fixed-length binary genomes and expresses phenotypes through interpretable rules inspired by biological principles such as dominance, recessivity, polygenic contribution, mutation, pleiotropy, genotype-dependent health burden, and environment-dependent performance. The term *biomimetic* is used here in a constrained methodological sense: biological systems provide design abstractions, not direct proof that the artificial system reproduces real biology. This interpretation is consistent with biomimicry as an interdisciplinary approach that studies and transfers principles or mechanisms from nature to design problems [10, 11]. PishieGen therefore does not claim to model a real species with empirical fidelity. Instead, it uses biological concepts as structured design constraints for an artificial-life simulator.

The initial organism template in this paper is a stylized small mammalian predator, because such organisms allow intuitive links between morphology, camouflage, thermoregulation, sensory acuity, locomotion, circadian behavior, and ecological niche. Some trait examples are motivated by domestic cat genetics, including coat-pattern development [12], short-limb variants [13], Manx-associated tail mutations [14], and Scottish Fold osteochondrodysplasia [15]. These examples are used only to define conservative computational analogues of inheritance constraints and trait trade-offs. The simulator is designed so that species templates can be replaced without changing the general modeling framework.

Contributions. This paper makes four contributions. First, it defines a compact digital genome and an auditable genotype–phenotype expression pipeline. Second, it formalizes inheritance, mutation, and health-constraint mechanisms for artificial organisms. Third, it defines environment-dependent fitness as a transparent multi-component function rather than as an opaque score. Fourth, it proposes a reproducible validation and experimentation plan aligned with best practices in individual-based and agent-based modeling, including model specification using the ODD protocol [5–7].

2 Related Work

2.1 Artificial life and open-ended evolutionary systems

Artificial life emerged as an interdisciplinary attempt to synthesize and study life-like processes in artificial media [1]. Its scope includes digital organisms, artificial chemistries, cellular automata, evolutionary algorithms, ecological simulations, self-organization, and embodied agents. A recurring question is whether artificial systems can support open-ended evolution: the sustained production of adaptive novelty, complexity, and new organizational forms [2, 3]. PishieGen is not presented as a full open-ended evolution system. Its narrower goal is to create an interpretable intermediate platform where inherited digital structure, phenotype expression, and environment-dependent selection pressure can be inspected and modified.

2.2 Genotype–phenotype maps

A genotype–phenotype map assigns genetic states to expressed traits, structures, or functions. In natural systems, the map is shaped by molecular biology, development, regulation, and

environmental interactions. In artificial systems, the map is designed, which permits experimental control but also risks oversimplification. The review by Manrubia et al. [4] emphasizes that genotype-space structure, phenotypic bias, robustness, and accessibility influence evolutionary dynamics. PishieGen translates this insight into software design by separating raw genome encoding from phenotype expression and by making the mapping inspectable at the level of individual loci, trait modules, and fitness components.

2.3 Individual-based and agent-based ecological modeling

Individual-based models represent populations as collections of distinct entities with state variables and behavior rules. They are widely used when heterogeneity, history, local interactions, and adaptive behavior matter [8]. Agent-based modeling provides a related abstraction for studying emergent dynamics in complex systems. Because such models can become difficult to reproduce, the ODD protocol was developed to standardize descriptions of individual-based and agent-based models [5]. Later updates improved clarity, replication, and structural realism [6, 7]. PishieGen follows this tradition by defining organism state variables, environmental state variables, scheduling, submodels, initialization, input parameters, and output metrics.

2.4 Biomimicry and design abstraction

Biomimicry studies biological forms, processes, and systems as sources of design principles [11]. Recent work frames biomimicry as a sustainable and interdisciplinary design methodology involving abstraction from biological observation to engineering implementation [10]. In PishieGen, biomimicry is not used as a rhetorical claim that artificial organisms are biologically real. Instead, it constrains model design. For example, thermal compatibility is inspired by thermoregulation, camouflage is inspired by background matching and disruptive coloration, and locomotor suitability is inspired by morphology–substrate interaction. These abstractions are then implemented as explicit computational functions.

3 Model Overview

PishieGen represents an artificial ecosystem as a discrete-time stochastic simulation. The system contains a population of organisms, a set of biome environments, and processes for phenotype expression, mating, mutation, health screening, movement or deployment across environments, and performance evaluation.

Let each organism i have a genome $g_i \in \{0, 1\}^L$, where $L = 128$ in the reference implementation. A phenotype expression function ϕ maps the genome and optional developmental noise ϵ to a phenotype vector:

$$x_i = \phi(g_i, \epsilon_i), \quad x_i \in \mathbb{R}^{d_x} \times \mathcal{C}^{d_c}, \quad (1)$$

where $\mathbb{R}^{d_x}$ denotes continuous trait dimensions such as heat tolerance or agility and $\mathcal{C}^{d_c}$ denotes categorical traits such as coat pattern or tail morphology. Each environment e has a state vector

$$z_e = (T_e, H_e, V_e, S_e, I_e, R_e, Q_e), \quad (2)$$

where T is temperature profile, H humidity, V vegetation/cover density, S substrate type, I illumination regime, R resource/prey availability, and Q environmental quality or disturbance level.

Fitness is modeled as a composite function:

$$F(i, e, t) = \sigma \left(\sum_{k=1}^K w_k f_k(x_i, z_e, t) - \lambda H_i \right), \quad (3)$$

where f_k are interpretable compatibility components, w_k are non-negative weights, H_i is health burden, λ controls the penalty of deleterious traits, t is simulation time, and σ is a bounded transformation such as the logistic function. This formulation avoids a single unexamined fitness score and allows each component to be inspected separately.

4 Genome Representation

The reference genome length is 128 bits. This size is sufficient for a compact proof-of-concept while remaining easy to inspect and serialize. The bit layout is not treated as a biological genome. It is a computational genotype with biologically inspired modules. Table 1 provides the default allocation.

Table 1: Reference 128-bit digital genome layout. The mapping is a modeling convention, not a claim about natural genomic architecture.

Bits	Module	Purpose
0–7	Primary pigmentation	Main color or pigment code used in phenotype expression.
8–15	Secondary pigmentation	Hidden or recessive pigmentation code available for inheritance.
16–19	Pattern class	Solid, striped, spotted, ticked, point-like, or mixed pattern categories.
20–23	Pattern intensity	Degree of contrast, white spotting, or pattern coverage.
24–27	Fur length/type	Short/long expression, carrier status, undercoat, curl, or hair-reduction analogue.
28–31	Ear morphology	Ear size and structural variants affecting hearing and heat exchange.
32–35	Tail morphology	Tail length, bobtail/tailless analogues, and balance-relevant variants.
36–39	Paw and extremity traits	Polydactyly analogue, grip, substrate stability, and extremity contrast.
40–47	Health loci	Deleterious or lethal genotype combinations; health-burden modifiers.
48–55	Thermal profile	Heat and cold tolerance components.
56–63	Camouflage profile	Background-matching and disruptive-pattern parameters.
64–71	Locomotor traits	Agility, sprint capacity, climbing, balance, and substrate response.
72–79	Sensory traits	Hearing, low-light perception, prey detection, and sensory noise.
80–87	Behavioral traits	Caution, exploration, learning rate, and risk sensitivity.
88–95	Circadian tendency	Diurnal, nocturnal, or crepuscular activity preference.
96–111	Mutation markers	Flags for mutation origin, structured mutation class, or experimental lineage tags.
112–127	Reserved	Versioning, species-template identifiers, and future experimental modules.

4.1 Allelic abstraction

For categorical traits, an allele is represented by an integer decoded from a bit segment. For traits with hidden variation, the primary and secondary segments are interpreted as a diploid-like pair:

$$a_{m,1} = \text{decode}(g_m^{(1)}), \quad a_{m,2} = \text{decode}(g_m^{(2)}), \quad (4)$$

where m indexes a trait module. The expressed phenotype may depend on dominance hierarchy, recessive equality, epistasis, or masking. For example, if $D(a)$ is a dominance rank, a simple expression rule is:

$$\phi_m(a_{m,1}, a_{m,2}) = \begin{cases} a_{m,1}, & D(a_{m,1}) > D(a_{m,2}), \\ a_{m,2}, & D(a_{m,2}) > D(a_{m,1}), \\ \psi_m(a_{m,1}, a_{m,2}), & D(a_{m,1}) = D(a_{m,2}), \end{cases} \quad (5)$$

where ψ_m handles codominance, blending, or recessive expression. This abstraction is intentionally simple, but it is sufficient to support Mendelian validation experiments.

4.2 Polygenic traits

Traits such as intelligence, agility, camouflage, and thermal tolerance are modeled as polygenic aggregates rather than single-locus categories. If B_m is the set of bit positions contributing to a trait m , a normalized additive model is:

$$y_m = \frac{1}{|B_m|} \sum_{b \in B_m} g_b. \quad (6)$$

The expressed continuous trait may then include nonlinear transformations and developmental noise:

$$x_m = \text{clip}(\alpha_m + \beta_m y_m + \eta_m, 0, 1), \quad \eta_m \sim \mathcal{N}(0, \sigma_m^2). \quad (7)$$

This permits heritability without deterministic cloning of parental performance.

5 Inheritance and Mutation

5.1 Mating operator

Given two parent genomes g^A and g^B , offspring generation proceeds module by module. For diploid-like modules, one allele is sampled from each parent:

$$a_{m,1}^C \sim \text{Uniform}\{a_{m,1}^A, a_{m,2}^A\}, \quad a_{m,2}^C \sim \text{Uniform}\{a_{m,1}^B, a_{m,2}^B\}. \quad (8)$$

For polygenic modules, each bit may be sampled from either parent or inherited through recombination blocks:

$$g_b^C = \begin{cases} g_b^A, & u_b < 0.5, \\ g_b^B, & u_b \geq 0.5, \end{cases} \quad u_b \sim \text{Uniform}(0, 1). \quad (9)$$

A recombination-block option can be introduced to preserve linkage between adjacent loci.

5.2 Mutation

Mutation is modeled at two levels. First, independent bit-flip mutation occurs with probability μ_b :

$$g_b^C \leftarrow 1 - g_b^C \quad \text{with probability } \mu_b. \quad (10)$$

Second, structured mutation rules may be defined for specific allele combinations:

$$(a_{m,1}, a_{m,2}) \rightarrow a_m' \quad \text{with probability } \mu_m(a_{m,1}, a_{m,2}). \quad (11)$$

Structured mutations should be documented in a public table, because undocumented mutation recipes would make the model difficult to reproduce.

5.3 Health constraints and non-viability

Some genotype combinations may be marked as non-viable or deleterious. This is not included to dramatize the simulation, but to represent a general biological fact: not all inherited combinations are equally compatible with development and function. In feline genetics, several traits provide well-documented examples of strong structural or health consequences, including Manx-associated variants in *T/Brachyury* [14], short-limb Munchkin-associated structural variation [13], and Scottish Fold-associated *TRPV4* variation [15]. In PishieGen, these are represented only as abstract health-constraint templates.

Let $\mathcal{L}$ be a set of lethal genotype patterns and $\mathcal{D}$ a set of deleterious genotype patterns. The viability function is:

$$V(g) = \begin{cases} 0, & g \in \mathcal{L}, \\ 1, & \text{otherwise.} \end{cases} \quad (12)$$

For viable organisms, health burden is:

$$H(g) = \sum_{r \in \mathcal{D}} \omega_r \mathbf{1}[g \text{ matches } r], \quad (13)$$

where ω_r is the severity weight of rule r . All lethal and deleterious rules must be explicitly documented.

6 Environment and Biomimetic Fitness

6.1 Biome state

Each biome is a structured object rather than a label. A desert, forest, tundra, wetland, or grassland differs through climate, substrate, vegetation, illumination, and resource availability. The simulator therefore avoids statements such as “desert-adapted” unless the underlying compatibility components are visible.

A biome state at time t is:

$$z_e(t) = \{T_e(t), H_e(t), C_e(t), S_e, L_e(t), R_e(t), P_e(t)\}, \quad (14)$$

where C is cover density, S substrate, L illumination, R resource/prey availability, and P predator or hazard pressure. Time-varying terms can be deterministic, stochastic, or empirically supplied.

6.2 Fitness components

The reference fitness function is decomposed into six components:

$$F = \sigma(w_T f_T + w_C f_C + w_L f_L + w_S f_S + w_B f_B + w_R f_R - \lambda H), \quad (15)$$

where f_T is thermal compatibility, f_C camouflage compatibility, f_L locomotor compatibility, f_S sensory compatibility, f_B behavioral timing, and f_R resource-handling efficiency.

Thermal compatibility can be expressed as a distance penalty between organism tolerance and environmental temperature:

$$f_T = \exp\left(-\frac{(T_e - T_i^*)^2}{2\sigma_T^2}\right), \quad (16)$$

where T_i^* is the organism's preferred thermal range center. Camouflage compatibility can be represented as similarity between phenotype color-pattern vector c_i and environmental background vector b_e :

$$f_C = 1 - \frac{\|c_i - b_e\|_2}{\sqrt{d_c}}. \quad (17)$$

Behavioral timing links circadian tendency to illumination and resource timing:

$$f_B = \text{sim}(q_i^{\text{circadian}}, q_e^{\text{resource}}(t)). \quad (18)$$

These equations are intentionally simple in the first implementation. The priority is interpretability and testability before biological realism.

7 Simulation Schedule

A single simulation step consists of:

1. update environmental variables such as illumination, temperature, and resources;
2. express phenotypes for newly generated or modified organisms;
3. assign organisms to environments according to an experimental policy;
4. compute component-level and aggregate fitness;
5. sample survival, resource acquisition, or reproductive eligibility events;
6. apply mating rules to selected pairs;
7. apply mutation and viability checks to offspring;
8. record population-level metrics.

This schedule can be implemented as a deterministic pipeline with seeded random number generation to support exact reproducibility.

8 Software Architecture

The proposed software architecture is modular:

- **genome:** binary encoding, decoding, recombination, mutation, and viability checks;
- **phenotype:** trait expression, dominance rules, polygenic aggregation, and trait metadata;

- **environment:** biome definitions, climate cycles, resource schedules, and background profiles;
- **fitness:** transparent component-level compatibility scoring;
- **simulation:** population management, breeding experiments, selection policies, and generation loops;
- **analysis:** diversity metrics, inheritance validation, sensitivity analysis, and visualization.

The implementation should include tests for each biological abstraction. For example, if two heterozygous carriers are crossed under a simple Mendelian rule, simulated offspring ratios should converge to the expected 1:2:1 genotype distribution as sample size increases. This type of test guards against implementation errors and makes the model auditable.

9 Validation and Experimental Plan

Agent-based and individual-based models require careful validation because they can produce plausible-looking dynamics even when assumptions are poorly specified. Recent reviews emphasize that validation can involve docking, empirical comparison, visualization, sensitivity analysis, and related methods [9]. PishieGen’s first validation layer is internal rather than empirical: the simulator should reproduce known consequences of its own declared rules.

9.1 Experiment 1: Mendelian inheritance check

Two parents with genotype Aa are crossed under a single-locus dominance model. The expected genotype distribution is:

$$P(AA) = 0.25, \quad P(Aa) = 0.50, \quad P(aa) = 0.25. \quad (19)$$

The simulator should reproduce these ratios within binomial sampling error over large N .

9.2 Experiment 2: Polygenic regression-to-parental-range check

For a polygenic trait such as agility or thermal tolerance, offspring values should correlate with parental values but retain variation. The experiment estimates parent–offspring correlation, trait variance, and response to selection across generations.

9.3 Experiment 3: Biome-specific fitness comparison

A controlled panel of organisms is evaluated across multiple biomes. The expected result is not that one organism is globally optimal, but that different trait profiles produce different environment-specific advantages. Output should report component scores, not only aggregate fitness.

9.4 Experiment 4: Selective breeding under environmental pressure

A population is repeatedly selected for high performance in one biome. The experiment tracks allele frequencies, phenotype means, health burden, and loss of diversity. This tests whether optimization for one environment creates trade-offs in others.

9.5 Experiment 5: Mutation-driven diversity

Mutation rate is varied across simulation runs. The outcome metrics include allelic richness, phenotype diversity, viability loss, and maximum observed fitness. This experiment identifies regimes where mutation is too low to generate novelty or too high to preserve functional organization.

10 Reproducibility

Every run should be specified by a configuration file containing genome layout version, mutation rates, dominance tables, health rules, environment parameters, selection policy, random seed, and output metrics. The repository should include:

- a machine-readable genome specification;
- documented dominance and mutation tables;
- unit tests for inheritance, mutation, viability, and fitness components;
- notebooks for each experiment;
- fixed random seeds for baseline results;
- continuous integration for test execution.

11 Limitations

PishieGen is an artificial model, not a biological model of real organisms. Several limitations follow. First, binary genomes are a computational convenience and do not represent real genomic architecture. Second, dominance and recessivity are simplified relative to real developmental genetics. Third, phenotype expression ignores many biological mechanisms, including gene regulation, epigenetics, developmental timing, microbiome effects, and social learning. Fourth, the fitness function is designer-specified and should not be interpreted as empirical ecological fitness. Fifth, feline trait examples are used as constrained analogues only; the model should avoid making veterinary or breeding recommendations.

These limitations are acceptable for an artificial-life simulator if they are explicitly stated and if experiments are interpreted as consequences of model assumptions rather than as direct biological findings.

12 Ethical and Scientific Scope

The project should not encourage harmful animal breeding practices. Traits associated with known health burdens should be treated as examples of why genotype constraints matter, not as desirable optimization targets. In the software interface, health-burden metrics should be visible and should penalize rather than reward deleterious combinations. The broader scientific value of the project lies in computational reasoning about inheritance, variation, adaptation, and trade-offs, not in aesthetic selection for extreme morphologies.

13 Conclusion

PishieGen proposes a reproducible biomimetic artificial-life simulator centered on explicit genotype–phenotype mapping and environment-dependent fitness. Its main design principle is interpretability: genomes, traits, inheritance, mutation, health constraints, and fitness components should be visible and testable. By combining artificial-life methodology, genotype–phenotype abstraction, individual-based ecological modeling, and biomimetic design principles, the simulator can serve as a research-oriented open-source platform for studying stylized adaptive dynamics in digital organisms. Future work will implement the reference Python package, publish baseline experiments, add sensitivity analysis, and compare alternative genotype–phenotype maps under controlled selection regimes.

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